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FROM
SUBJECT COLLECTEUR LUNAIRE

Message

Dear Humbert Camerlo,

Please find attached a number of sketches showing a design for a prototype 'Collecteur lunaire'.

The size of the collecteur is approximately 1.4m x 1.4m, and will focus on an area about 5.5 m from the centre of the collecteur. The size of this 'focus area' is theoretically about 250mm x 250mm, but an area of 400mm x 400mm can be realistically expected given manufacturing tolerances. The theoretical magnification factor of the collecteur is 25:1.

If you wish to make a larger or smaller collecteur than the DIMENSIONS can simply be scaled up or down.

This prototype can be used for any full moon when the altitude of the moon is less than 40 degrees. This is true for all full moons in April, May, June, July and August.

I also enclose some details of the movements of the moon for the full moons in July and August.

We will send you some more details of how to set up the collecteur later this week.

Yours sincerely,

Andrew Sedgwick

Andrew Sedgwick

If you have not received all the pages listed please phone us on the above number.

PROTOTYPE 'COLLECTEUR LUNAIRE' - BRIEF DESCRIPTION

Schedule of pieces :

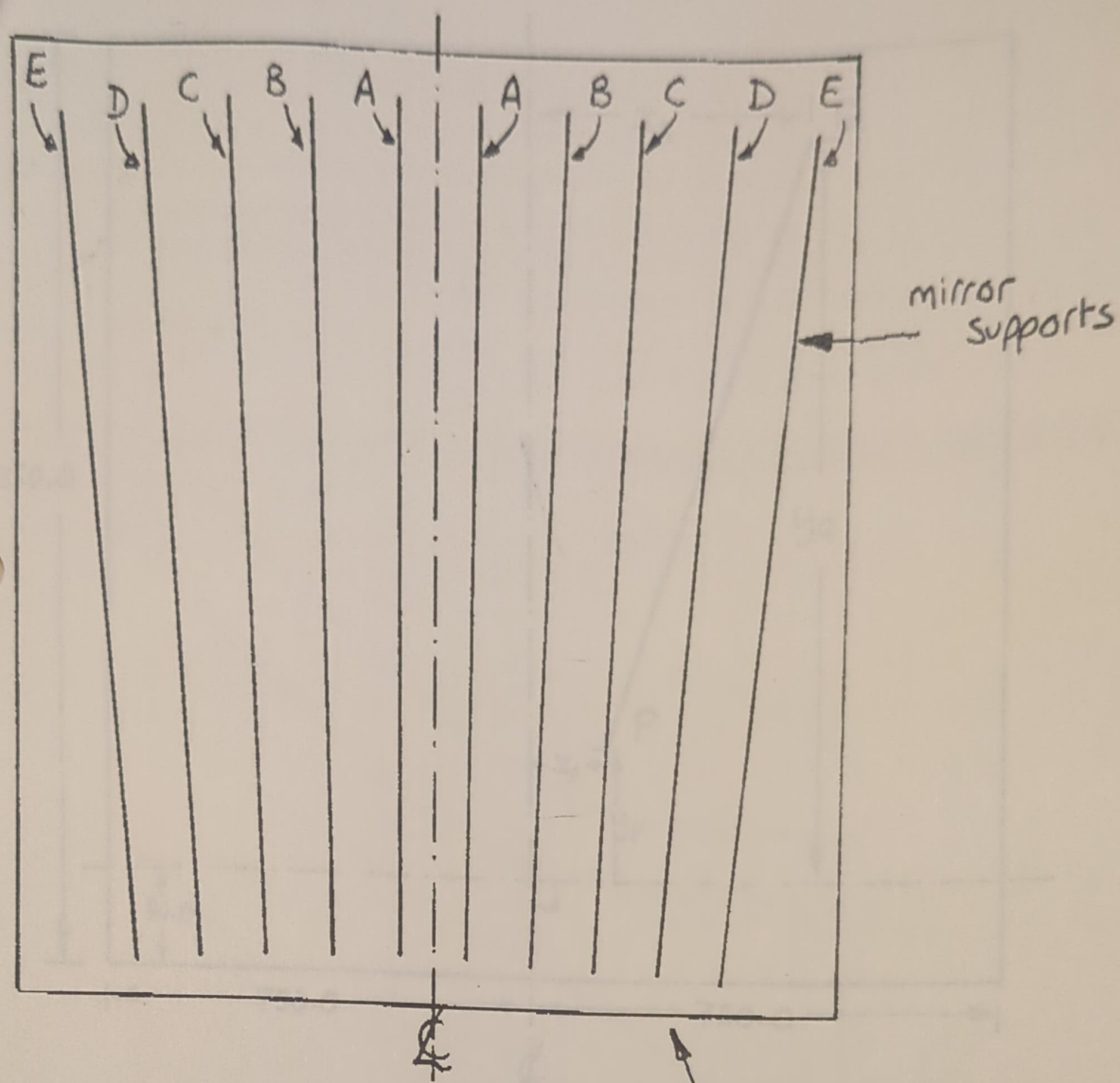
- 1 x wooden baseboard, 1500mm x 1350mm
- 10 x wooden mirror support pieces, each 1245 mm long with varying heights (see schedule on sheet 7)
- 25 x square flat mirrors each about 220mm x220mm
- 1 x aiming tube with cross-hairs (see sheet 8)

Description :

Ten mirror support pieces are mounted vertically on the wooden baseboard. The exact positioning of the support pieces on the baseboard is given in the schedule on sheet 4. The layout of the support pieces is symmetrical about the centreline of the baseboard - there are two type 'A' supports, two type 'B' supports ... etc.

Each square mirror rests on two supports. Each pair of supports carries five mirrors - see sheet 5. It is important that the the mirrors are placed centrally on the supports so that the centre of the mirror falls midway between the supports. It does not matter that there are gaps between the mirrors.

An aiming tube is mounted on edge of the baseboard so that the collecteur can be accurately aligned with the moon. The aiming tube is fixed at an angle of 65.5 degrees to the baseboard - see sheet 8. It does not matter where on edge of the baseboard the tube is fixed - but the position of the viewer for aiming should be considered.

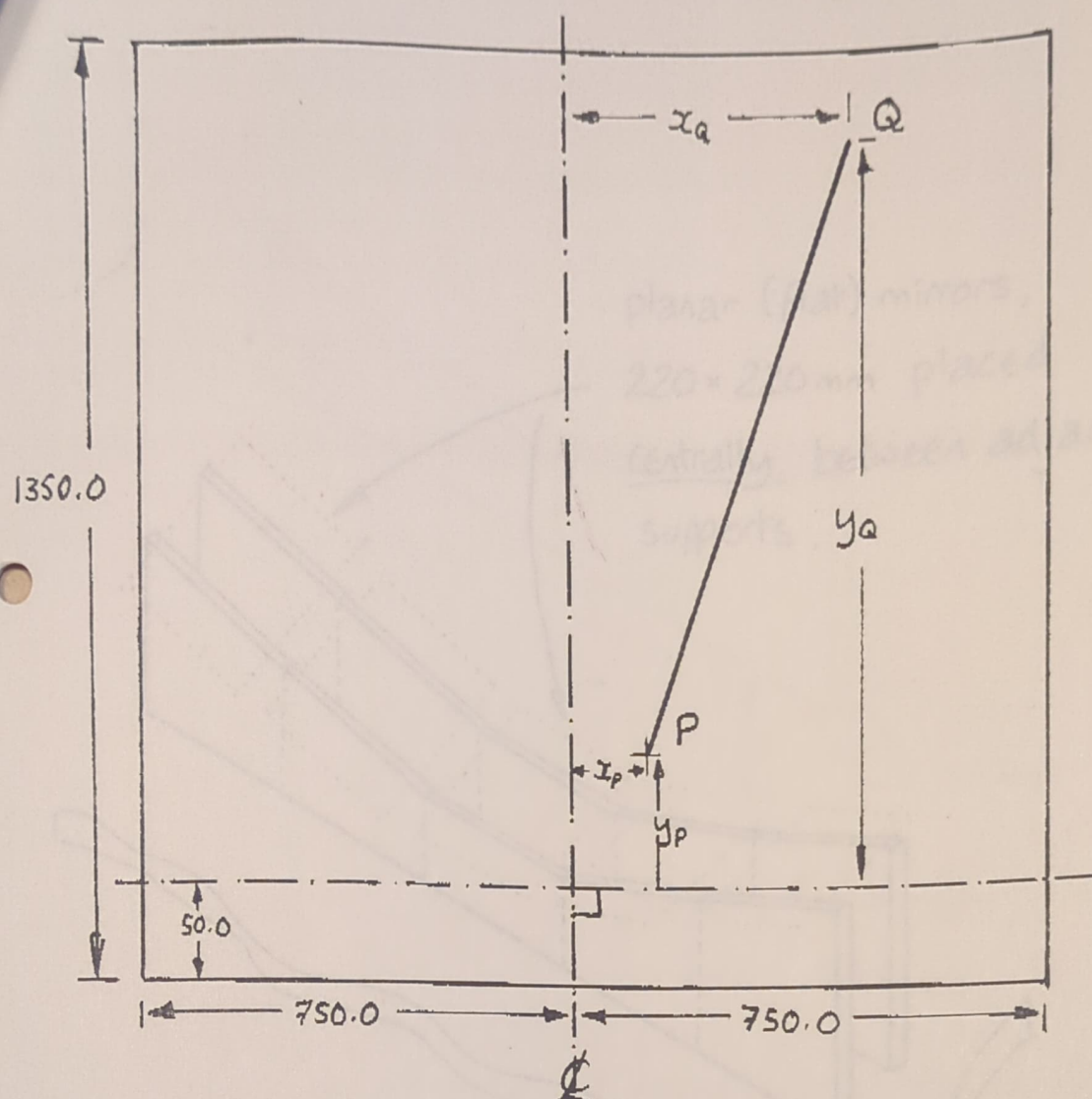


Symmetrical
about
centre line

baseboard

PLAN

(4)

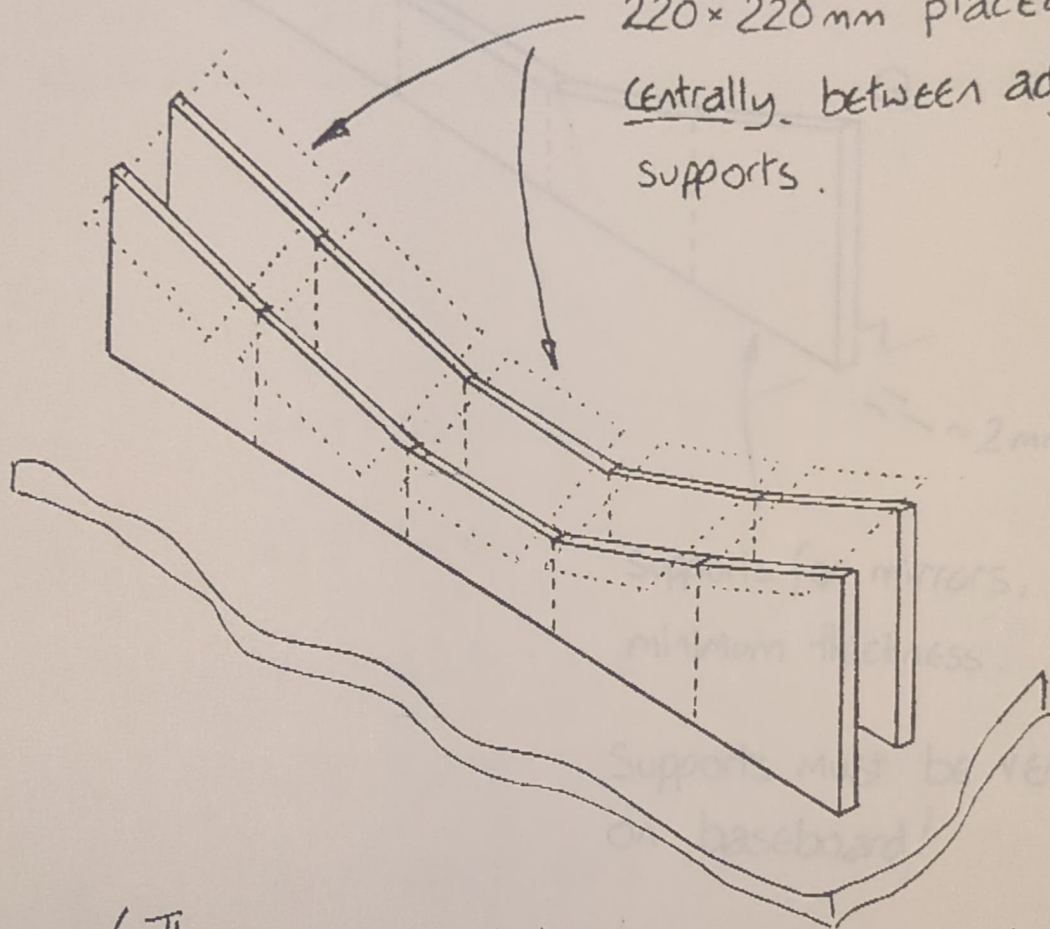
NOT TO SCALE!

PIECE	x_p	y_p	x_q	y_q
A	54.7	33.6	70.2	1278.9
B	164.0	30.9	210.4	1275.4
C	273.3	25.4	350.7	1268.4
D	382.2	17.3	490.5	1258.0
E	491.0	6.4	630.1	1244.0

note: Symmetrical about centre line

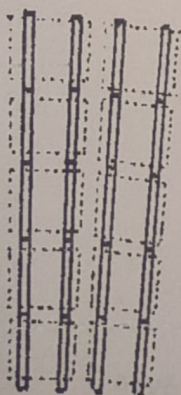
5

planar (flat) mirrors,
220×220 mm placed
centrally between adjacent
supports.



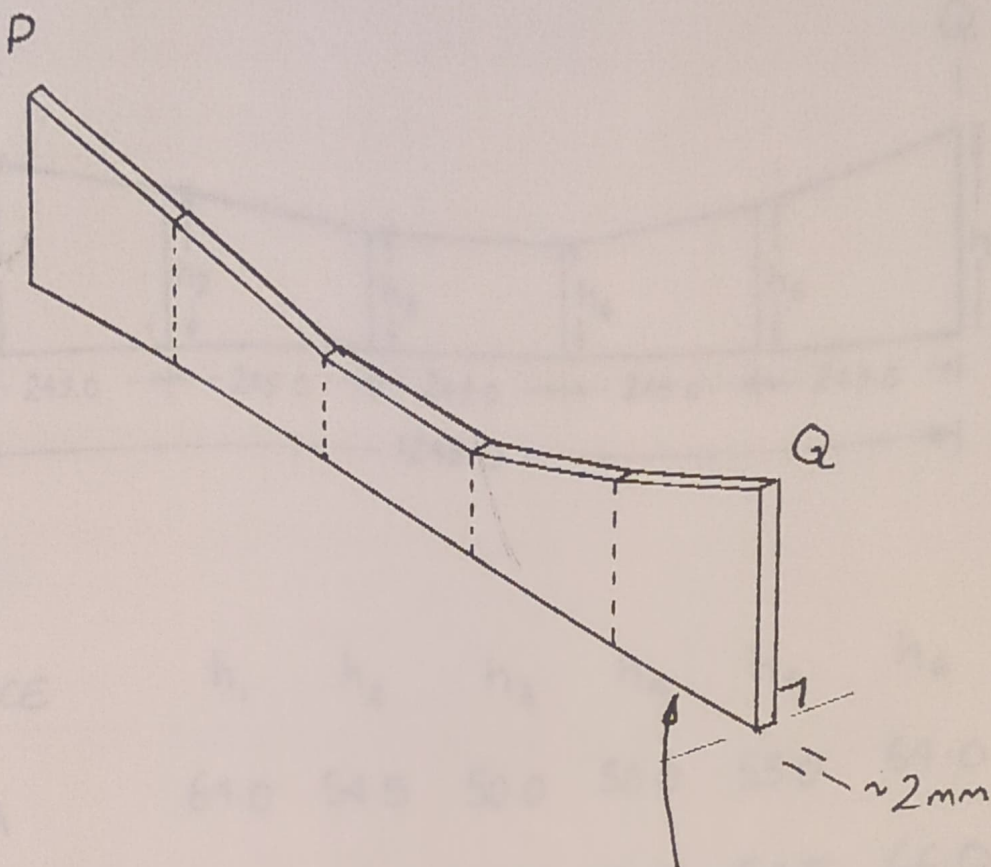
(There are meant to be gaps between
the mirrors !)

PART PLAN →
(not to scale !)



NOTE:

25 mirrors in total.



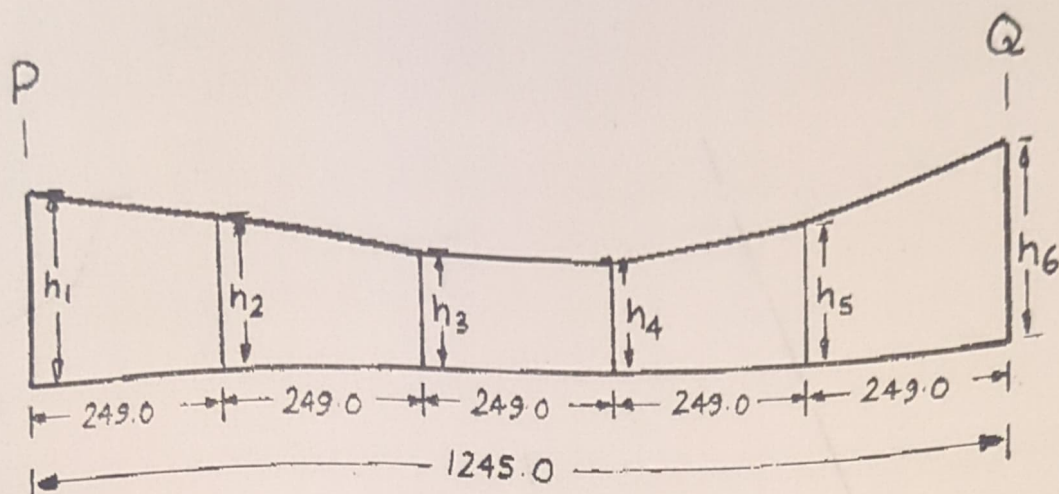
supports for mirrors,
minimum thickness.

Supports must be vertical
on baseboard!

All dimensions in millimeters

NOT TO SCALE!

(7)

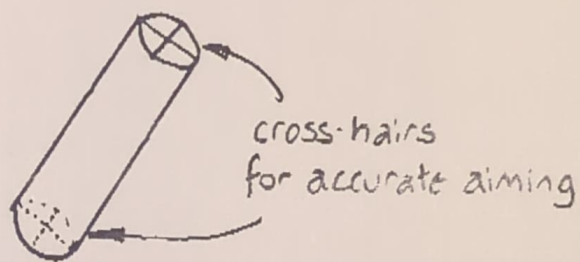
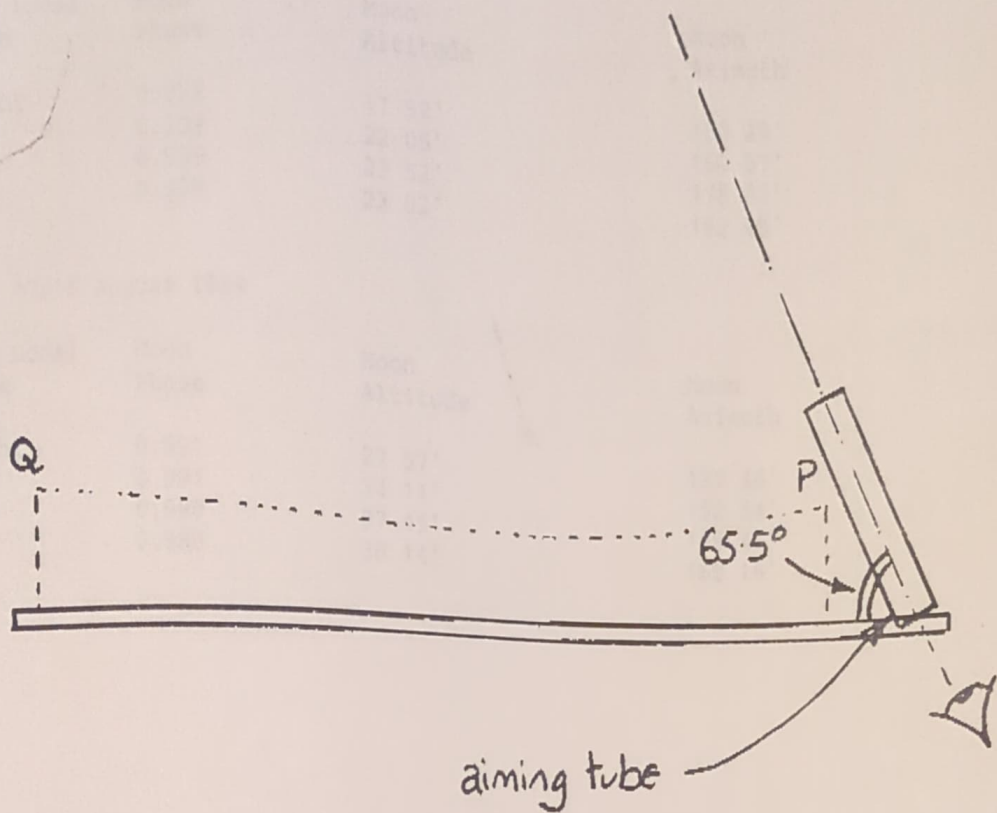


PIECE	h_1	h_2	h_3	h_4	h_5	h_6
A	64.0	54.5	50.0	50.0	55.0	64.0
B	65.5	56.0	51.5	51.5	56.5	66.0
C	68.0	58.5	54.0	54.5	59.5	69.0
D	71.5	62.5	58.5	59.0	64.0	73.5
E	76.5	68.0	64.0	64.5	70.0	80.0

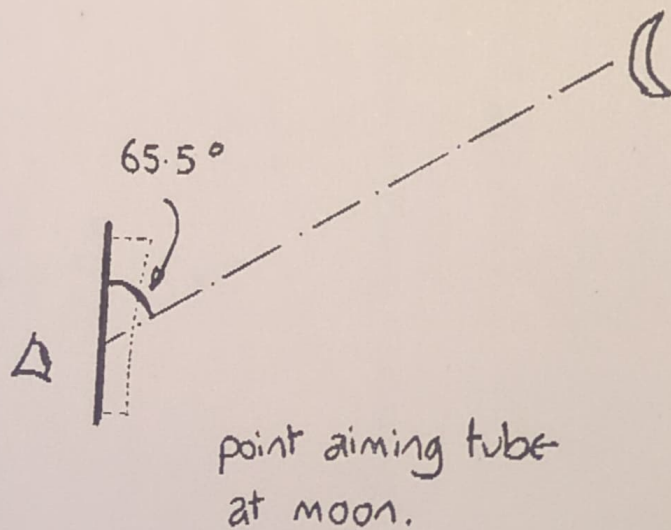
All dimensions in millimetres

NOT TO SCALE !

(8)



USE OF COLLECTEUR →



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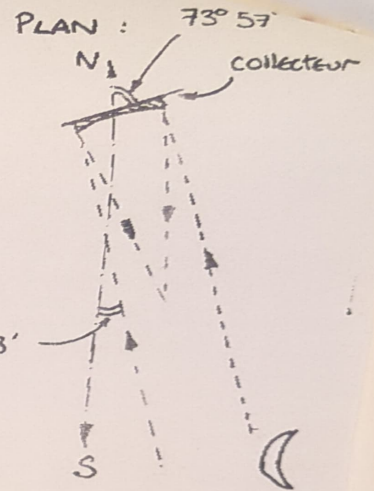
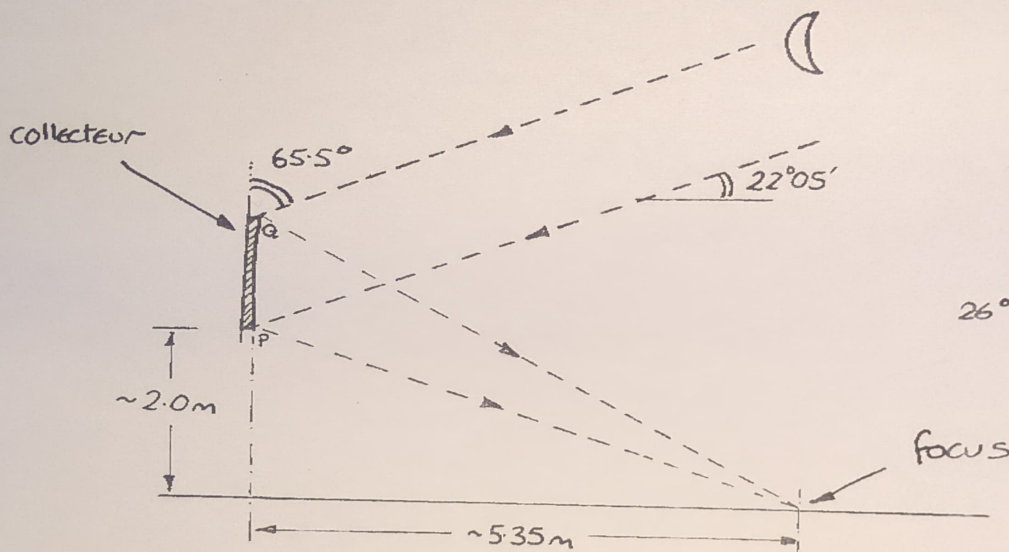
Data on Full Moons, July and August 1989

Date : 18/19 July 1989

France Local Time	Moon Phase	Moon Altitude	Moon Azimuth
midnight	0.999	17 52'	
01:00	0.999	22 05'	150 29'
02:00	0.999	23 52'	163 57'
03:00	0.998	23 02'	178 02'
			192 48'

Date : 17/18 August 1989

France Local Time	Moon Phase	Moon Altitude	Moon Azimuth
midnight	0.991	27 57'	
01:00	0.991	34 11'	137 46'
02:00	0.990	37 45'	152 54'
03:00	0.988	38 14'	169 45'
			188 16'



SECTION THROUGH MOON
AZIMUTH PLANE
(163°57' EAST OF NORTH)

Set-up for July 19th
01:00